Bulbul Singh

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SUMMARY

Final-year Computer Science student and software developer who builds from real problems. Java for problem solving, the MERN stack for web work, and hands-on experience with RAG and AI-powered applications. Two internships, national-level results and team leadership. Seeking software development internships and full-time roles.

EDUCATION

B.Tech, Computer Science | *ABES Engineering College, Ghaziabad, Uttar Pradesh* 2023 – 2027

Class XII | *Bijnor Public School, Bijnor, Uttar Pradesh* | 92.40% 2023

Class X | *St. Mary's School, Bijnor, Uttar Pradesh* | 95.83% 2021

SKILLS

Programming Languages: Java, C, JavaScript, TypeScript, SQL, Python

Databases: PostgreSQL (pgvector), MongoDB, MySQL, Prisma

Core Computer Science: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering

Web Technologies: HTML5, CSS, React.js, Next.js, Node.js, Express.js, Tailwind CSS

AI and APIs: RAG, vector embeddings, Cohere, Gemini (Genkit), REST, GraphQL, JWT, OAuth

Tools and Deployment: Git, GitHub, Vercel, Render, Neon

EXPERIENCE

Specialist Programmer L1 | *Infosys* Incoming

JSON Configuration Intern | *Hunt Digital Media* Jul 2026 – Sep 2026

- Translated human-readable scripts into JSON configurations that drive workflow-based interfaces: multi-stage flows, input fields, selectable options and action links.
- Supported conditional flows and data-capture fields, and contributed to the team's script-to-JSON conversion tool.

Workflow Automation Intern | *GEMA Education Technology Pvt Ltd* Sep 2025 – Dec 2025

- Supported workflow automation for programme outreach to schools and organisations, running templated, personalised email campaigns with Mailmeteor and mail-merge tools.
- Maintained the structured contact and programme data these workflows depended on.

PROJECTS

GitHub Knowledge Assistant *AI Codebase Q&A (RAG)* Live demo Code

Tech Stack: Next.js, TypeScript, Node.js, Express.js, PostgreSQL (pgvector), Prisma, Cohere

- Built a full-stack RAG application that indexes any public GitHub repository and answers questions in plain English, citing the exact files and line ranges used.
- Designed the pipeline: overlapping 40-line chunking, Cohere embeddings, pgvector similarity search and a prompt that restricts answers to the retrieved code.
- Handled API rate limits with sub-batching and exponential backoff; deployed on Vercel, Render and Neon.

LoopBack *Spaced-Repetition Tracker for LeetCode* Live app Code

Tech Stack: Node.js, Express.js, PostgreSQL, JWT, Passport.js, JavaScript, Render, Neon

- Built and deployed a multi-user web app with JWT authentication (Google OAuth implemented), bcrypt hashing, rate limiting and per-user data isolation in PostgreSQL.
- Integrated LeetCode's GraphQL API with retry and backoff and idempotent inserts; designed interval-based revision scheduling.

YieldX *AI-Powered Smart Farming Platform | Smart India Hackathon 2025* Live app Code

Tech Stack: Next.js, TypeScript, Tailwind CSS, MongoDB Atlas, JWT, Genkit (Gemini), Recharts

- Led the team and built the frontend of a multilingual (English, Hindi, Punjabi) decision-support app for farmers: an interactive dashboard with yield, profitability and sustainability views, plus text-to-speech recommendations.
- Collaborated on AI-powered crop diagnosis, a farming chatbot and user authentication.

POSITIONS OF RESPONSIBILITY

Webmaster | *ABES ACM-W Chapter* 2023 – Jul 2026

- Volunteer (2023), team member (2024), webmaster (May 2025 – Jul 2026); managed the chapter website and ran web development sessions.
- During my tenure, our chapter became the first featured in the ACM-W global newsletter spotlight, where I was named and quoted.

Finance Intern | *E-Cell ABESEC* Oct 2023 – Jul 2026

- Maintained financial records, budgets and event documentation; assisted with budgeting and sponsorship tracking by compiling and validating information from multiple sources.

Organising Team | *ACPC, ABES ACM and ACM-W Chapters* 2023, 2025

- Headed ACPC 2k25 and was a team member for ACPC 2k23, the ICPC-style inter-college programming contest of the chapters.
- Judged debates, open mics, group discussions and poetry and writing events.

ACHIEVEMENTS AND CERTIFICATES

- All India Rank 56, National Entrepreneurship Challenge 2025 (Advanced track, IIT Bombay)
- 2nd position, ARDETHON hackathon
- Google AI Essentials (Google)
- Top 15 teams, Western Entrepreneurship Challenge 2025
- Top 50 of 500+ teams, college round, Smart India Hackathon 2025
- SQL Intermediate (HackerRank)